

Migration Report: IIS → Kestrel + Caddy

MCS Employee API | July 13, 2026

Summary

The MCS Employee API (ASP.NET Core 8 Web API) was moved off IIS / IIS Express. It is now self-hosted on **Kestrel**, ASP.NET Core's built-in web server, with **Caddy v2.11.4** in front of it as a reverse proxy. The Windows IIS services (W3SVC and WAS) were stopped and set to Manual start so they no longer occupy port 80. All endpoints and the Swagger UI were verified working through the new stack.

New architecture

```
Client ----> Caddy (:80, reverse proxy) ----> Kestrel (http://localhost:5000)
              Caddyfile (repo root)              deploy\api\MCS app.dll
```

Changes made

Item	Change
MCS app/Program.cs	Added ForwardedHeaders middleware (X-Forwarded-For / X-Forwarded-Proto from Caddy); removed UseHttpsRedirection (TLS is now the proxy's job); Swagger enabled in all environments so it is reachable through Caddy.
MCS app/appsettings.Production.json	New file. Binds Kestrel to http://localhost:5000 in Production.
Caddyfile (new, repo root)	Caddy listens on :80 and reverse-proxies to localhost:5000; JSON access log written to logs/caddy-access.log.
deploy/api	Fresh Release publish output (dotnet publish -c Release).
start-servers.ps1 / stop-servers.ps1 (new)	Convenience scripts that start/stop Kestrel and Caddy together.
README.md	Documented the new hosting model and run instructions.
IIS (system)	W3SVC and WAS services stopped and startup type set to Manual, freeing port 80.
Caddy (system)	Installed via winget (CaddyServer.Caddy, user scope), v2.11.4.

Verification

- GET http://localhost/api/employees through Caddy returned **200 OK** with seeded employee data.
- GET http://localhost/swagger/index.html through Caddy returned **200 OK**.
- Port 80 is owned by **caddy.exe**; port 5000 by **dotnet.exe** (Kestrel, Production environment).
- W3SVC and WAS are **Stopped** (StartType: Manual); no IIS process holds port 80.

How to run

```
.\start-servers.ps1    # starts Kestrel (localhost:5000) + Caddy (:80)
.\stop-servers.ps1     # stops both
```

```
API base URL:  http://localhost/api/employees
Swagger UI:    http://localhost/swagger
```

Notes / next steps

- IIS was stopped, not uninstalled. To restore it: `Start-Service WAS, W3SVC` (elevated).
- Kestrel and Caddy currently run as background processes; for production-grade resilience, register both as Windows services (e.g. with NSSM or `sc . exe`) or a scheduled task at boot.
- To serve HTTPS with a real domain, change the Caddyfile site address from `:80` to the domain name - Caddy will provision TLS certificates automatically.